nature communications

Article

https://doi.org/10.1038/s41467-025-56701-4

Big-data empowered traffic signal control could reduce urban carbon emission

Received: 14 June 2024

Accepted: 27 January 2025

Published online: 27 February 2025

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Urban congestion is a pressing challenge, driving up emissions and compromising transport efficiency. Advances in big-data collection and processing now enable adaptive traffic signals, offering a promising strategy for congestion mitigation. In our study of China's 100 most congested cities, big-data empowered adaptive traffic signals reduced peak-hour trip times by 11% and off-peak by 8%, yielding an estimated annual CO_2 reduction of 31.73 million tonnes. Despite an annual implementation cost of US$1.48 billion, societal benefits—including CO_2 reduction, time savings, and fuel efficiency—amount to US$31.82 billion. Widespread adoption will require enhanced data collection and processing systems, underscoring the need for policy and technological development. Our findings highlight the transformative potential of big-data-driven adaptive systems to alleviate congestion and promote urban sustainability.

Road traffic is a major global source of carbon emissions, generating approximately 5.88 billion tonnes of greenhouse gas (GHG) annually (measured in CO_2-equivalent), or 11.9% of total emissions, ranking third among all sectors[1]. Urban congestion further exacerbates these emissions. In cities like London, congestion alone contributes around 2.2 million tonnes (Mt) of CO_2 annually, equating to 15% of total road traffic emissions, which would require a pine forest the size of Sydney (12,000 km²) to offset[2]. Similar congestion impacts are observed in Paris (2.8 Mt, 13%), Berlin (0.42 Mt, 10.5%), and Amsterdam (0.06 Mt, 7%)[2]. The situation is particularly acute in rapidly growing megacities in China, where congestion-related emissions are amplified by high traffic demand and infrastructure constraints. In China, transport emissions surged from 96 Mt in 1990 to 996 Mt in 2021 - a compound annual growth rate of 7.83% driven by urbanization and economic growth[3]. For cities like Suzhou, Chengdu, and Harbin, reducing congestion could potentially cut emissions by 0.95 (12.7%), 1.53 (14.2%), and 0.98 Mt (12.9%) of CO_2, respectively[4]. This rise in urban traffic emissions poses a challenge to China's 2060 carbon neutrality target[5] and threatens progress on sustainable development goals (SDGs) related to sustainable cities, infrastructure, and climate action[6].

Efforts to mitigate urban congestion and transport emission can be classified under the "Avoid-Shift-Improve" (ASI) framework[7]. The "Avoid" strategies aim to reduce travel demand through compact urban planning and policies like congestion pricing and traffic restrictions[8–10]. The "Shift" strategies encourage a shift from private vehicles to sustainable modes such as public transit, cycling and walking, often supported by transit-oriented development (TOD) initiatives[11,12]. While both "Avoid" and "Shift" strategies require considerable investment in infrastructure or policy changes, which can be time- and cost-intensive, the "Improve" strategies focus on enhancing the efficiency of existing transport systems, often offering immediate congestion relief. Adaptive traffic signal control exemplifies an "Improve" strategy, optimizing traffic flow by dynamically adjusting signal timings and reducing the need for expanded road capacity. Recent advances in big-data collection and processing techniques have enabled cities to implement adaptive signals on a large scale[13,14], making real-time adjustments possible and enhancing overall traffic management.

Research on adaptive traffic signal control has progressed over decades, evolving from early heuristic methods to data-intensive reinforcement learning approaches[15,16]. Adaptive control methods fall

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https://doi.org/10.1038/s41467-025-56701-4

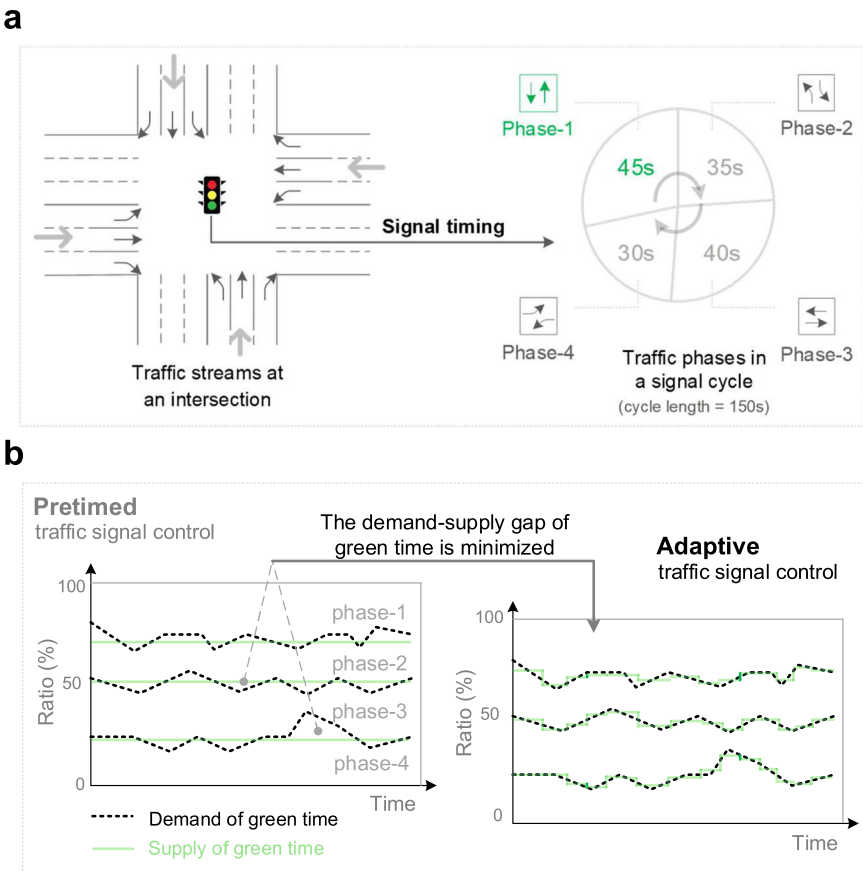

Fig. 1 | Comparison between pretimed and adaptive traffic signal control. a For a typical four-leg signalized intersection, specified traffic phases are designed to provide right-of-way for pairs of non-conflicting traffic streams. For example, in phase-1, southbound and northbound through movements are allowable. Phase timing is essential for improving intersection efficiency in each signal cycle. **b** With the pretimed method for phase timing, the lengths of green times for traffic phases remain fixed over a period of time (the horizontal solid lines). By contrast, the adaptive method continuously adjusts green times based on real-time traffic volume data. Thus, the gap between demand and supply of green time (or ratio) for each phase could be minimized.

into two primary categories: cycle-based methods (e.g., Webster's algorithm[17–19], SCATS, and SCOOT[20,21]) and phase-based methods (e.g., max-pressure[22–24] and reinforcement learning algorithms[25–27]). Cycle-based methods, which maintain consistent phase settings, are widely used because they require less data and provide straightforward rules for drivers to follow. Phase-based methods, which adjust phases in real time, offer greater adaptability but face challenges in Chinese cities where timely, high-quality data are often limited, and local driving and pedestrian behaviors require careful consideration. Although multi-source traffic data is now available, collecting and processing this data remains costly and can sometimes be unreliable. Therefore, signal timing update methods that are less data-dependent and more robust to data errors are still essential. In this context, it is not the novelty of traffic control algorithms but rather their compatibility with available data that becomes crucial. Consequently, simpler cycle-based algorithms like Webster's method have gained traction in China, particularly when integrated with platforms such as City Brain[28,29]. Pilot projects in Hangzhou and Nanchang have shown over 15% reductions in trip delays[30], demonstrating adaptive control's potential in lifting long-standing traffic restrictions without increasing congestion.

While pilot projects and extensive research on isolated intersections[31,32] show promise, systematic studies on the city-wide impacts of adaptive traffic signals - particularly considering both congestion and emissions reduction - are still limited. To address this, our study evaluates the large-scale implementation of big-data-driven adaptive signals in megacities, specifically focusing on urban areas with populations exceeding 10 million and thousands of signalized intersections. We address this research gap by conducting simulations in China's 100 most congested cities, using OpenStreetMap[33] for road data and Gaode Map for traffic speed data. Our analysis quantifies congestion and emission reductions at both isolated intersection and city-wide levels, identifies key socioeconomic and infrastructure factors influencing congestion mitigation, and assesses the cost-effectiveness of adaptive signal systems. Through this comprehensive examination of city-scale adaptive traffic signals, our study advances the understanding of smart traffic management and offers insights for developing low-emission, efficient urban mobility solutions aligned with China's sustainability targets.

Results

How adaptive traffic signal mitigates urban congestion?

To illustrate the potential of adaptive signals in reducing congestion and emissions, we begin by analyzing a standard four-leg intersection. At such intersections, each traffic phase is defined by green and yellow intervals allocated to non-conflicting traffic streams within a signal cycle (Fig. 1a). Efficient phase timing requires precise green time allocation and cycle length optimization across phases. Conventional pre-timed signals, which assign fixed green times based on historical averages (e.g., hourly or peak-period data), lack adaptability to real-time fluctuations in traffic demand. In contrast, adaptive signals dynamically adjust green times according to current traffic conditions, reducing mismatches between green time allocation and actual traffic needs (Fig. 1b). This flexibility suggests that citywide implementation of adaptive signals could significantly improve traffic flow and alleviate congestion.

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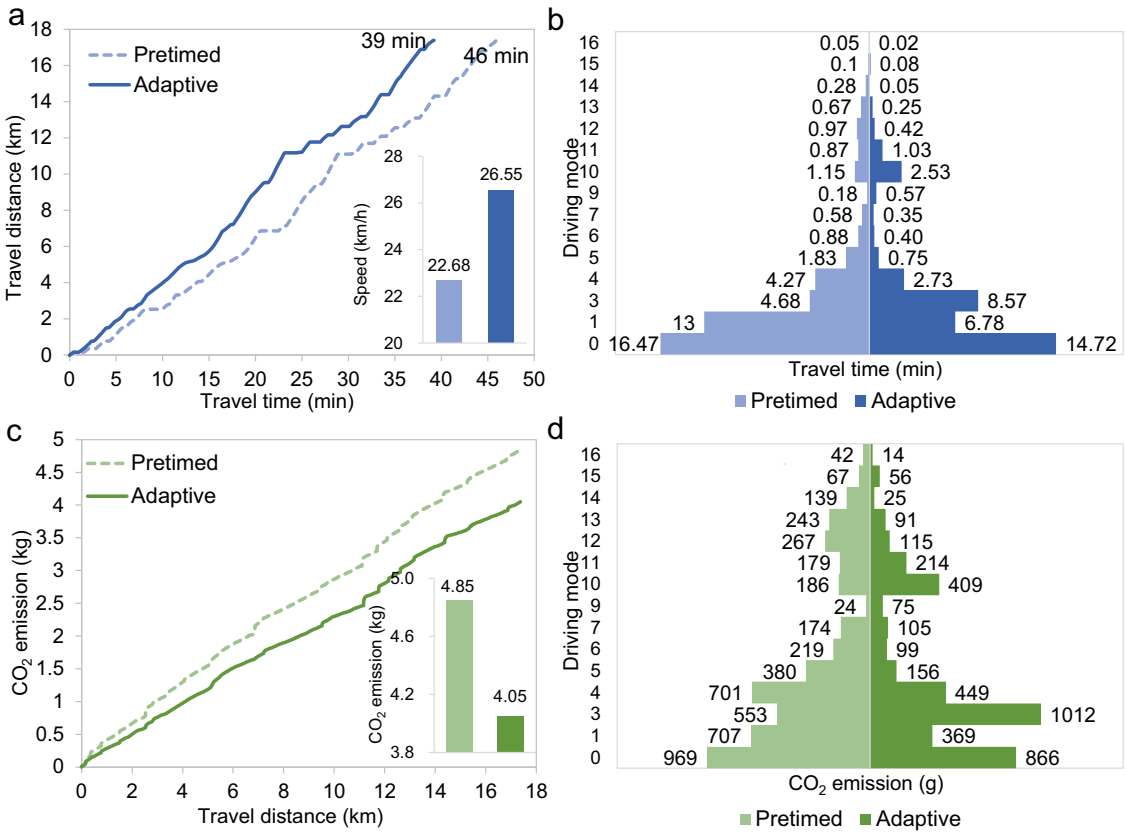

Fig. 2 | Impact of adaptive traffic signals on vehicle's travel speed, driving modes, and CO_2 emissions. a In Shanghai, the demo vehicle's travel speed increased from 22.68 to 26.55 km/h with adaptive traffic signals. **b** Comparison of durations in various driving modes for the demo trip: adaptive signals reduce time spent in braking (mode-0) and idling (mode-1) at intersections, with a notable increase in cruising time (mode-3 and mode-10). **c** CO_2 emissions decreased from 4.85 kg to 4.05 kg during the demo trip (16% reduction). **d** Comparison of CO_2 emissions across driving modes shows that emission savings from reduced braking and idling outweigh the increase during cruising, resulting in a net 16% emission reduction. Refer to Supplementary Table 1 for detailed driving mode definitions.

Adaptive traffic signals also contribute to emissions reduction by minimizing vehicle stops and idling time. In densely populated Chinese cities, the frequent proximity of intersections - averaging about 500 meters apart in urban centers - means that vehicles encounter multiple stops during typical trips. Furthermore, the complexity of urban traffic, including interactions with pedestrians and cyclists, often leads to additional braking and idling at intersections, exacerbating emissions and reducing traffic efficiency. This study directly compares pre-timed and adaptive signals in terms of their ability to smooth traffic flow and reduce emission, controlling for signal timing by applying Webster's method to both. The key distinction lies in data input: pre-timed signals rely on static historical data, while adaptive signals use real-time data to dynamically adjust green times. This dynamic adjustment reduces vehicle stops and decelerations, thereby lowering congestion and emissions.

To quantify these advantages, we conducted a demonstration trip spanning 17.37 km and 20 signalized intersections in Shanghai. Using the Motor Vehicle Emission Simulator (MOVES) model[34], we assessed adaptive signals' impact on driving patterns and CO_2 emissions. Adaptive control significantly decreased braking and idling times, reducing idling from 13 min to 6.78 min and braking time from 16.47 to 14.72 min, effectively halving idling duration. This reduction, coupled with an increase in cruising time under low- and medium-speed conditions, raised average travel speed by 17% and reduced trip duration by 7 min.

These improvements in driving patterns yielded notable emissions reductions. Along the demonstration route, CO_2 emissions fell from 4.85 kg to 4.05 kg, driven by the reduction in braking and idling. Overall, adaptive signals achieved a net CO_2 reduction of 16% (Fig. 2b), demonstrating that optimized signal timing not only improves traffic flow but also mitigates emissions by reducing stops and idling at intersections.

To evaluate the broader system-wide impact of adaptive signals, we conducted simulation-based experiments to analyze the relationship between system performance and varying levels of adaptive signal deployment. Using a business-as-usual (BAU) scenario with all intersections pre-timed as a baseline, we incrementally increased the proportion of adaptive signals across Shanghai. Results show that both travel speed and trip time reductions increased logarithmically with the percentage of adaptive implementation, for both peak and off-peak hours (Supplementary Fig. 1). Notably, deploying adaptive control at just the first 20% of intersections reduced peak-hour trip times by 8%, with diminishing returns observed beyond this point. This finding aligns with prioritization strategies where high-traffic intersections yield the greatest benefits[35]. Thus, cities aiming to implement adaptive signals for optimal congestion relief should prioritize intersections with the highest traffic volumes.

Congestion mitigation potential across China

Our study underscores that congestion mitigation potential is widespread across Chinese urban centers, extending well beyond major cities like Shanghai. Leveraging data from 100 most congested cities in China as ranked by Gaode in 2022, we simulated the transition from pretimed to adaptive traffic signals, observing substantial impacts. Notably, 97 cities exhibited meaningful increases in average speed

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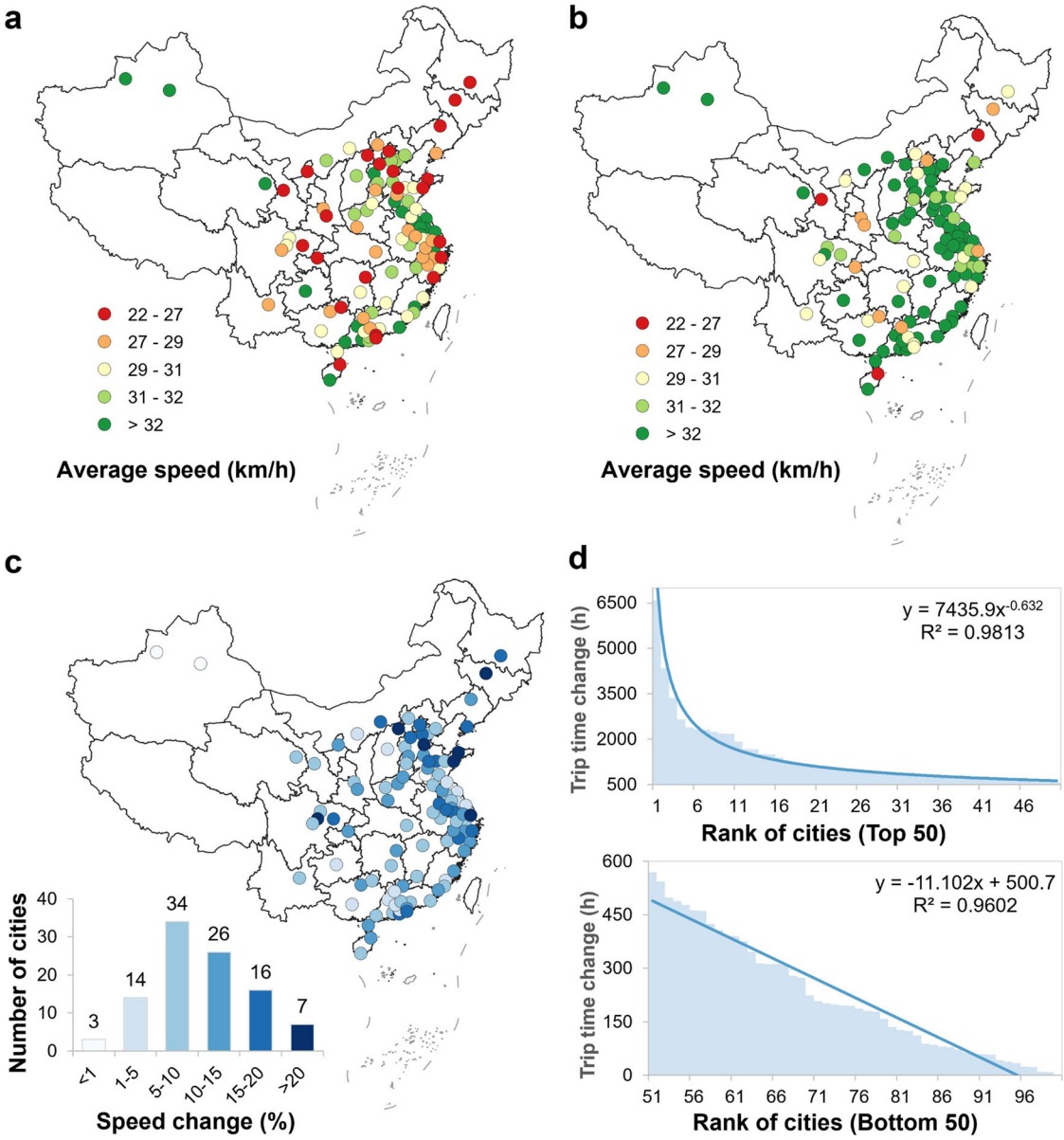

Fig. 3 | Changes of travel speed and trip time during peak hours. Geographic distribution of average speed under BAU scenario (all traffic signals are pretimed) (**a**), target scenario (all traffic signals are adaptive) (**b**), and the percent change (**c**). **d** Rank of cities by trip time reductions each hour, for top 50 cities, trip time reduction is fitted as a power function of rank; for bottom 50 cities, trip time reduction is fitted as a linear function of rank. The base map is applied without endorsement from GADM data (https://gadm.org/).

(exceeding 1%), with 49 cities achieving peak-hour speed gains of over 10%. Collectively, these improvements yielded an 11% reduction in trip time during peak hours (equivalent to 79,695 h saved) and an 8% reduction during off-peak hours (42,080 h saved) (Supplementary Fig. 2). For cities like Hangzhou and Nanchang, which have previously reported the integration of big data to enhance signal adaptivity, our baseline scenario (BAU) accounted for their existing adaptive traffic signals. Spatially, the highest mitigation potential appeared concentrated in coastal and provincial capital cities, regions integral to China's economic network (Fig. 3a–c). Ranking these cities by trip time reduction, we found Shanghai, Beijing, Harbin, Qingdao, and Shenzhen as the top five, with Shanghai alone achieving a notable reduction of up to 6581 h during peak hours, representing an 18% decrease and an average savings of 9 min per trip (Fig. 3d).

Interestingly, our analysis revealed a truncated power relationship between trip time change and city rank. Specifically, we observed a power function for the top 50 cities and a linear function for the bottom 50 cities (Fig. 3d). This finding suggests that the potential for congestion mitigation diminishes rapidly for high-ranked cities but more slowly for low-ranked cities. For example, the trip time reduction in Beijing (4339 h) would be 66% of that in Shanghai, whereas the difference between Shenzhen and Shenyang (ranked 5th and 10th, respectively) is only 10%. We hypothesize that the differences among low-ranked cities are primarily due to variations in travel demand influenced by socioeconomic factors such as population. In contrast, for high-ranked cities, the sharp differences in trip time reduction are jointly determined by both travel demand and speed change.

To explore the infrastructure and socioeconomic determinants of congestion mitigation potential in 100 cities, we conducted a multivariate linear regression analysis (Table 1). Our findings reveal that baseline BAU speed, road length and population negatively impact speed improvement. Specifically, a 1% increase in baseline speed, logarithms of road length and population corresponded to reductions of approximately 1.1%, 3.3%, and 2.3% in potential speed improvement,

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respectively. This is consistent with the expectation that higher baseline speeds leave less room for improvement and that higher population and road length, indicating greater travel demand towards downtown areas, make it more challenging to mitigate traffic congestion. In contrast, cities with more straight streets and higher gross domestic product based on purchasing power parity (GDP (PPP)) exhibited greater potential for speed improvement. In summary, our results suggest that implementing the adaptive method would be most beneficial in cities with high congestion levels, high economic level, and well-designed streets.

Table 1 | Regression analysis between speed improvement and city indicators

	Coefficient	*P*
BAU speed (km/h)	−1.129	<0.001
Log[road length (m)]	−3.330	0.076
Log[Population]	−2.295	0.298
Street straightness	106.770	<0.001
Log[GDP (PPP)]	2.113	0.079
Pseudo/adjusted R^2	0.486	
Method	OLS	

This table presents the result of a multivariate linear regression analysis of speed improvement (%) against baseline BAU speed, road network length, population, street straightness (ratio of straight-line distances between adjacent street nodes to street lengths)[39], and GDP (PPP). Negative and positive coefficients indicate that city indicators have negative and positive impacts on congestion mitigation potential, respectively. The *P*-value indicates the significance of each city indicator; the smaller the value, the more significant it is. The method used for linear regression is ordinary least squares (OLS).

Reduction of carbon emission

The annual road traffic carbon dioxide (CO_2) emissions across the 100 cities were approximately 477 Mt in 2021, according to the Climate Watch Database[3]. Our analysis reveals that implementing adaptive traffic signals could reduce CO_2 emissions by 31.73 Mt annually—a 6.65% reduction across these cities (Fig. 4). This reduction is equivalent to the offset that would be achieved by replacing 9.16 million (3.30%) petrol vehicles with electric vehicles each year in China (see *Methods*). The CO_2 reduction potential varied significantly by city, from 16 Kt in Sanya to 1.67 Mt in Shanghai (Fig. 4a), with the top 10 cities accounting for nearly 30% of the total emission reductions, while the bottom 10 cities contributed only 1.64%. Even lower-ranked cities, such as Changsha, which saw a 5.09% reduction, showed meaningful potential for emission reduction when considering percentage improvements in urban air quality and health outcomes.

Transport emissions reductions are influenced by both baseline emissions and the impact of adaptive signals. In 2021, baseline BAU CO_2 emissions across the cities ranged widely, from 0.5 Mt in Sanya to 20.6 Mt in Chongqing. In general, cities with higher BAU emissions show greater absolute reduction potential; however, reductions are not directly proportional to BAU emissions. For example, despite

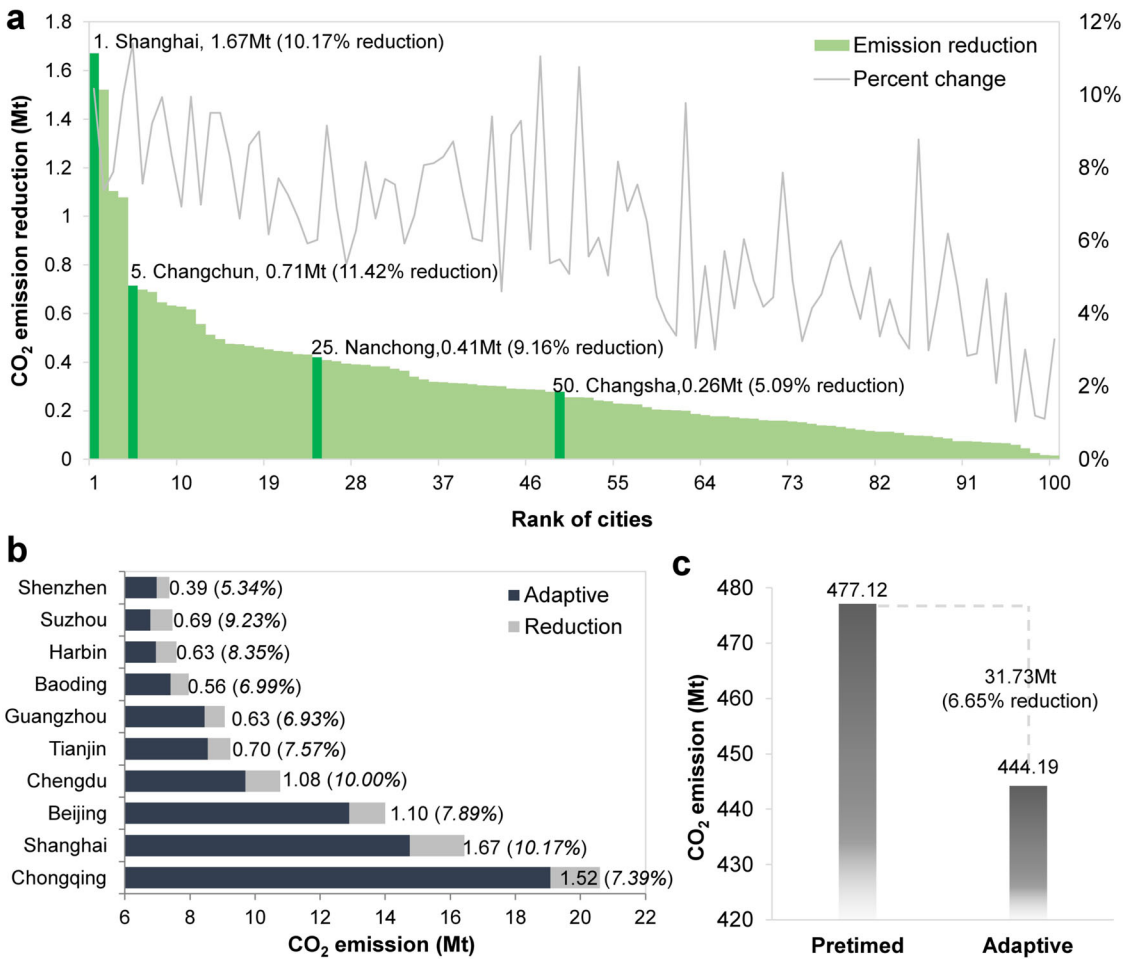

Fig. 4 | Reduction of CO_2 emission. a Histogram of annual CO_2 reductions with percentage changes indicated by a broken line, labeled for typical cities. **b** Ranking of the top 10 cities by BAU annual CO_2 emissions: dark bars represent predicted total emissions, while light bars show potential reductions under adaptive traffic signals, labeled are emission reduction and percent change. **c** Annual transport CO_2 emission reductions across 100 cities.

Shanghai's lower BAU emissions than Chongqing (Fig. 4b), Shanghai's reduction potential was higher (1.67 Mt vs. 1.52 Mt). This underscores the importance for cities to evaluate their specific reduction potential prior to implementation, as city-specific characteristics can significantly affect outcomes.

Assuming stable trip demand and route choices, emission reductions are primarily achieved by minimizing braking and idling durations and increasing individual vehicle travel speeds. In congested urban areas, inefficient driving modes and reduced speeds contribute heavily to emissions. According to the MOVES model, even moderate improvements in these driving modes could lead to substantial reductions. Beyond CO_2, the adaptive signals can also mitigate emissions of air pollutants like NO_x, NH_3, and VOCs, which are linked to PM2.5 and O_3 pollution. Thus, adaptive traffic signals contribute not only to CO_2 reduction but also to public health improvements by reducing pollutant exposure. This positions adaptive traffic control as a key strategy in China's pursuit of carbon neutrality and its alignment with the Sustainable Development Goals.

Cost and benefit

To effectively mitigate congestion and emissions, adaptive traffic signals require a comprehensive evaluation of both implementation costs and resulting benefits (Fig. 5a–d). The most substantial cost component - real-time traffic detection– demands continuous traffic volume data, which incurs considerable expense. Our estimates suggest that achieving the targeted congestion mitigation across 100 cities would necessitate a total investment of approximately US$1.48 billion, covering initial setup and ongoing maintenance (Fig. 5f). Specifically, based on an analysis of 145 recent field cases, the initial installation cost for traffic detectors is projected at US$1.31 billion, assuming a 71-month lifespan per detector[36]. Additionally, annual operation and maintenance costs are estimated at around US$0.17 billion. Unsurprisingly, larger cities bear higher costs due to the scale of intersections and network density.

Integrating multiple existing data sources can possibly reduce implementation costs. Using a combination of inductive-loop detectors, surveillance cameras, and floating car data allows for data substitution at intersections lacking traffic detectors (Supplementary Table 2). Yet, deploying these solutions is often complex due to varied ownership of traffic detectors and signals across public and private entities. We classify multi-source data integration solutions into three tiers: (1) Departmental solutions with low barriers using single-department data; (2) Cross-sector solutions requiring interdepartmental collaboration and policy support; and (3) Public-private partnerships, offering broader data coverage at higher implementation barriers. Higher-tier solutions promise less traffic detection costs but entail coordination costs difficult to quantify. Centralized smart city systems, such as City Brain, highlight the potential of cross-sector data-sharing to lower detection implementation costs while supporting networked governance[37] for targeted deployment from high- to low-traffic areas (Supplementary Fig. 4).

The societal benefits of adaptive traffic signals are substantial, primarily from reductions in travel time and fuel consumption. Based on stated preference (SP) survey data of travelers' willingness to pay for reduced travel times in China[38], time savings are valued at approximately US$1.56 billion annually. Increased travel speeds further contribute to reductions in fuel consumption, yielding an estimated US$15.83 billion in annual fuel savings (Fig. 5f). Additionally, adaptive signals generate environmental benefits, with CO_2 emissions reductions valued at approximately US$14.44 billion annually. In addition to CO_2, adaptive signals help mitigate air pollutants such as NO_x, NH_3, and VOCs, which contribute to PM2.5 and O_3 pollution, as well as traffic accidents and noise. The full scope of these environmental benefits is challenging to quantify accurately, meaning the actual value could be even higher. Nevertheless, our estimates align with China's carbon-neutral policy priorities, reinforcing the relevance of adaptive traffic signals in environmental policy. Collectively, adaptive signals can deliver societal and environmental benefits of approximately US$31.82 billion per year across the 100 cities, representing a benefit-cost ratio of about 21.6. This compelling ratio underscores adaptive traffic signals as a highly effective strategy for urban congestion mitigation and environmental improvement.

Individual city assessments reveal notable variability, with benefit-cost ratios exceeding 30 in half of the 100 cities, and nearly a quarter showing ratios above 50 (Fig. 5e). These high ratios are largely driven by fuel savings and CO_2 reductions, while travel time benefits and implementation costs are relatively consistent across cities. Cities such as Urumqi and Yili, however, may not experience immediate net cost benefits from travel-time savings alone, with a benefit-cost ratio below 1/3 (Supplementary Fig. 3). Yet, when broader societal and environmental gains are factored in, these cities see overall benefits exceeding costs by more than 20 times, primarily due to emissions reductions and fuel savings. These findings indicate that cities focused narrowly on congestion alleviation may benefit from adopting a more comprehensive view, as our study highlights the substantial societal and environmental co-benefits of adaptive signals.

Despite these promising outcomes, many pilot projects in China—such as those in Hangzhou and Nanchang—still concentrate predominantly on congestion reduction, often overlooking broader environmental and fuel-saving benefits. This gap arises partly due to the complexities of estimating city-wide emissions reductions. Our analysis provides critical insights into these co-benefits, underscoring the strategic potential of adaptive traffic signals to support urban sustainability and inform policy adjustments for broader implementation across Chinese megacities.

Discussion

China's urbanization rate is projected to rise from 62.5% in 2021 to 80% by 2050, with increased migration to mega-cities posing both challenges for congestion and opportunities for decarbonization. Advances in data collection and processing now make adaptive traffic signals a practical solution for managing congestion and reducing urban emissions. This study introduces a systematic framework for modeling and assessing the benefits of big-data-driven adaptive traffic signals at a city-wide scale, particularly in megacities with populations exceeding 10 million. Our findings indicate that, while travel time reductions and implementation costs of adaptive signals are often similar, the emissions reduction benefits tend to outweigh the costs. Even in smaller cities such as Sanya, Yili, and Urumqi, where gains in travel speed and emissions may be more modest, our study provides valuable insights. These cities can apply our framework to evaluate the broader societal and environmental advantages of adaptive traffic signals, even potentially informing policy shifts to leverage big data for targeted transport emission reduction goals.

Our benefit-cost analysis may understate adaptive signals' value due to limitations in data availability. First, implementation costs could be overestimated; cities in China increasingly utilize floating car data from multiple sources from navigation and ride-hailing apps (e.g., Gaode Map, Didi), potentially reducing installation costs of traffic detectors at intersections with high floating car penetration. However, lacking comprehensive data, we could not quantify this impact. Second, we did not account for co-benefits such as noise reduction and accident mitigation associated with smoother traffic flow. Despite these limitations, our analysis provides a robust reference for policymakers, with opportunities to refine benefit-cost accuracy as more comprehensive data become available. Importantly, our findings support the adoption of adaptive traffic signals as a valuable tool for urban sustainability and emission reduction, even with conservative estimates.

With road transport contributing 11.9% of global CO_2 emissions and ranking among the top three emitting sectors worldwide, data-

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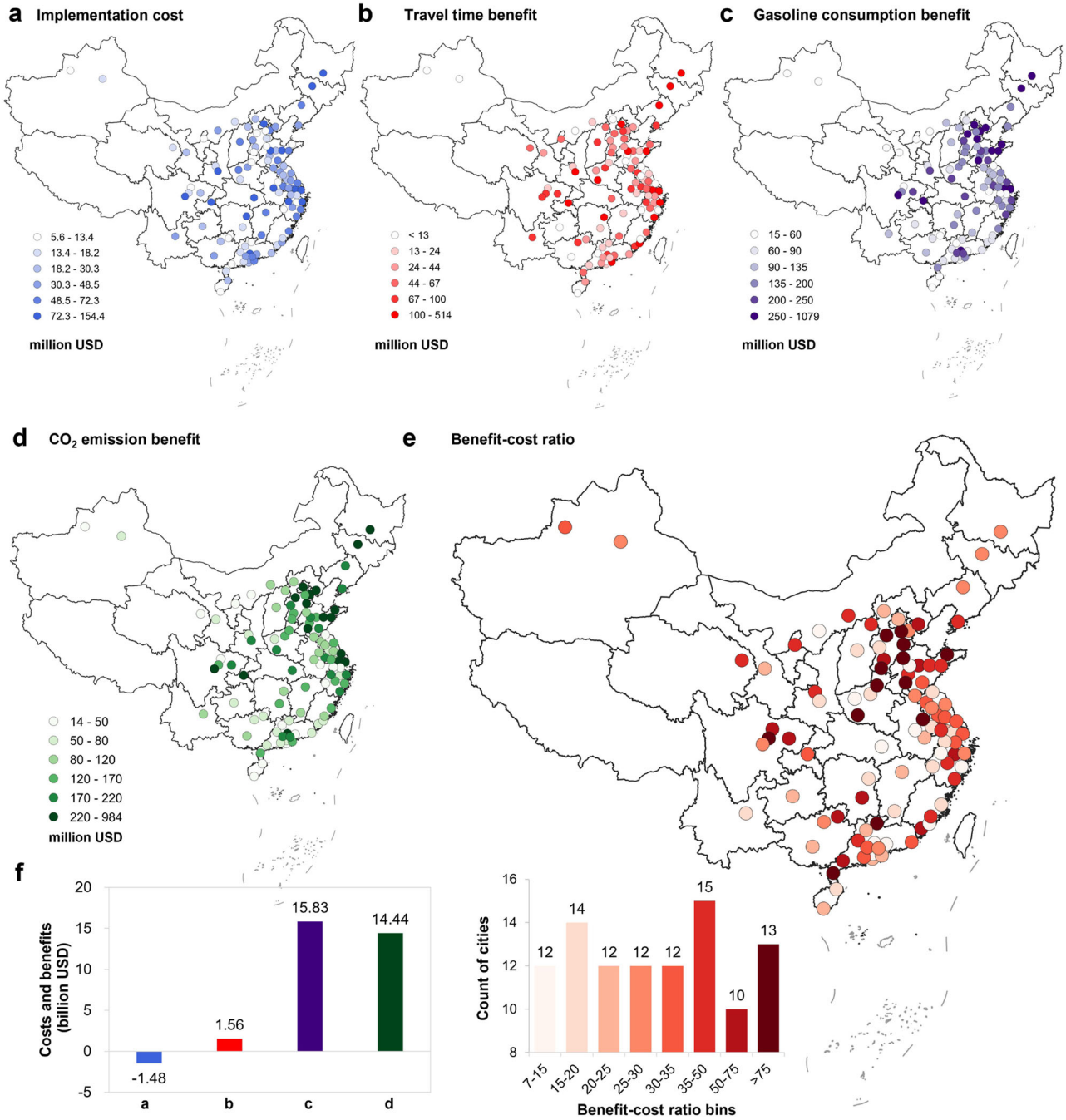

Fig. 5 | Costs and benefits of implementing adaptive traffic signal control. Geographic distribution of costs and benefits across cities: implementation cost (including installation and maintenance of traffic detectors) (**a**) benefits from reduced travel time (**b**), reduced gasoline consumption (**c**), and reduced CO_2 emissions (**d**). **e**, Benefit-cost ratios across 100 cities. **f**, Net costs and benefits in 100 cities, with negative values representing costs and positive values representing benefits. The base map is applied without endorsement from GADM data (https://gadm.org/).

driven congestion mitigation solutions offer notable potential for supporting China's carbon neutrality and sustainable development goals (SDGs). Future research could expand on our framework by examining the emission reduction potential of integrated "Avoid-Shift-Improve" solutions. For instance, further studies could explore "Improve" measures like dynamic variable lanes and green wave coordination, "Shift" strategies such as transit-oriented development, and "Avoid" approaches like the 15-min city concept. Our framework and models provide a foundation for assessing city-wide congestion and emissions reduction strategies, aligning with broader efforts toward urban sustainability.

Methods

Data sources

We sourced road network data from OpenStreetMap and adjusted urban boundary errors for cities like Guangzhou and Hangzhou (https://www.openstreetmap.org/). Urban boundaries, population, and GDP data were retrieved from the GHS Urban Centre Database (available at https://ghsl.jrc.ec.europa.eu/ghs_stat_ucdb2015mt_r2019a.php). The GHS Urban Centre Database (GHS-UCDB) describes spatial entities called "urban centers" according to a set of multi-temporal thematic attributes gathered from the GHSL sources integrated with other sources available in the open scientific domain. The

Urban Centres are defined by specific cut-off values on resident population and built-up surface share in a 1 × 1 km uniform global grid. Traffic data, including speed and congestion levels for China's top 100 cities, were collected from Gaode (https://report.amap.com/diagnosis/index.do) for the period from November 2 to November 8, 2022. Information on pilot projects in Hangzhou and Nanchang was obtained from online reports and studies[28,30].

Baseline BAU transport CO_2 emission data for 2021 were drawn from the Climate Watch Database (https://www.climatewatchdata.org/ghg-emissions). To estimate the gasoline consumption, we used the result of CO_2 emission from gasoline from the Engineering Toolbox (available at https://www.engineeringtoolbox.com/co2-emission-fuels-d_1085.html). The price of gasoline is retrieved from Trading Economics on 2022/11/08 (available at https://tradingeconomics.com/commodity/gasoline). We collected data from 145 field projects that were recorded in the Intelligent Transportation Systems Joint Program to estimate installation and maintenance costs of adaptive traffic signals at intersections (available at https://www.itskrs.its.dot.gov/).

Experiment setup

We used a series of simulation-based experiments to evaluate the performance of pretimed and adaptive traffic signal timing methods. Road network query, trip generation, and simulation environment calibration are the three main tasks for experiment setup (Supplementary Fig 5a).

Road network query. We used the OSMnx[39] package to query road networks given a polygon boundary. When querying road networks, we set the "network_type" as the "drive" option to include drivable public streets (but not service roads) for vehicles in a city. The road network was then divided into 8 × 6 identical traffic zones for generating synthetic trips.

Synthetic trip generation. We used a two-step procedure to generate synthetic trips within a city downtown area: (1) trip demand estimation, and (2) trip route assignment. We used the gravity model[40,41] to estimate the trip demands (T) between pairwise traffic zones over the simulation period. A uniform random choice model was used to select intersection locations as origin and destination for each trip. With such a model, each intersection within a specified traffic zone was selected at an equal probability.

$$T = \frac{m_i^{\alpha} \cdot m_j^{\beta}}{\gamma \cdot r_{ij}^{\delta}} \tag{1}$$

where m denotes the population of a traffic zone. Since the raw population data is at city-scale, we estimated m by multiplying overall population with the proportion of road length in a traffic zone. The subscripts i, j are used to denote the origin and destination traffic zones, respectively. And r denotes Euclidean distance between center-points of origin and destination traffic zones. The parameters $\alpha, \beta, \gamma, \delta$ are estimated for each city at the simulation environment calibration step.

To assign trips (T) to travel routes, we first divide the overall simulation period into 10-min intervals. In each interval, we use the expected travel time based shortest algorithm (Dijkstra's algorithm) for trip assignment[42]. The BPR function[43,44] is used to estimate the expected travel time (tt) of a road segment.

$$tt = tt_f \cdot \left[1 + 0.15 \cdot \left(\frac{x}{1000}\right)^4\right] \tag{2}$$

where, tt_f is the time cost for passing a road segment when traveling at speed limit. And x denotes the assigned trips to a road segment in previous time intervals, at beginning of simulation period, x is set to be 0.

Simulation environment calibration. We input the road network and generated synthetic trips into CBEngine[45] for simulating traffic in a city. CBEngine is a traffic simulator that allows for city-scale traffic simulation. To narrow the gap between simulation and real-world, we used a trial-and-error procedure to iteratively adjust trip demand parameters (i.e., parameter-$\alpha, \beta, \gamma, \delta$) to minimize the difference between observed and simulated average speed at city-scale (Supplementary Fig 5b). The observed speed data were retrieved from Gaode in 2022 during peak and off-peak hours.

Traffic signal timing. We used traffic volume data to compute the two signal timing variables: (1) cycle length and (2) length of green interval for each traffic phase. We used the typical phase setting as shown in Fig. 1a for all four-leg intersections. For three-leg intersections, one of the four signal phases is dropped based on the intersection configuration. Here, we used Webster's method[1,17,18] to calculate signal cycle length. This method requires estimation of saturation flow rates, critical flow ratios and lost times of each traffic phase. Here, we use subscript i to denote a traffic phase. For any traffic phase, saturation flow rate (s_i) is estimated as the maximum volume passed an unsignalized highway, averaging over 10 rounds of simulation. To compute critical flow ratio (y_i), we need to identify the critical traffic stream with maximum observed volume (q_i), in a traffic phase. The critical flow ratio is then calculated as follows,

$$y_i = \frac{q_i}{s_i} \tag{3}$$

In each traffic phase, a portion of beginning of each green interval (start-up lost time) and a portion of each yellow interval (clearance lost time) is not usable by vehicles, the sum of these two periods compromises the lost time l_i for a phase. Here, the lost time for each phase is assume to be 5 s. Using critical flow ratios and lost times, the signal cycle length is computed as below.

$$C = \frac{1.5L + 5}{1 - Y} \tag{4}$$

where $Y = \sum_i^n y_i$ and $L = \sum_i^n l_i$ denote the summed critical flow ratios and lost times over all traffic phases in a signal cycle, respectively. And the superscript n denotes total number of phases in a cycle, in general, $n = 4$ for typical four-leg intersections.

Next, the length of green interval for each traffic phase is computed as below, and the yellow interval is assumed to be 3 s for all phases.

$$g_i = (C - 3n) \cdot \frac{y_i}{Y} \tag{5}$$

It is worth noting that the pretimed and adaptive methods differ in signal timing update frequency and the data used for update. Here, our proposed adaptive method uses real-time volume data collected in the previous signal cycle to update signal timings for the coming cycle. By contrast, the pretimed method uses historical volume data to update signal timing variables every 15 min. The historical volume data are generated by the same road network simulation but with variations on trip departure times.

Scenario design. To evaluate the performance of the proposed adaptive method, we designed a set of baseline BAU and target traffic scenarios for 100 cities. In the BAU scenario, all traffic signals in a city's downtown area were pretimed. The preset traffic signal timings were derived from synthetic historical volume data using Webster's method. The synthetic historical traffic volumes were comparable to target scenario traffic but with minor differences due to randomness in departure time choice of specified trips.

In our target scenarios, we implemented partial or all intersection signals using our proposed adaptive method. To explore congestion mitigation potential in a city, we used a target scenario with all adaptive traffic signals. To study the impact of implementation rate, we ranked all intersections by total traffic volumes of all traffic streams over an hour and selected intersections from high to low volume ones. We assume that traffic management agencies would follow such a quasi-optimal implementation order since high volume intersections are often more critical.

Urban congestion. We used total trip time, average travel speed, CO_2 emission reduction and fuel consumption savings to evaluate the benefits of the proposed adaptive method. Trip time and travel speed were selected considering both drivers' and traffic police's perspectives, respectively. We computed average travel speed and total trip time by querying trip trajectories from simulator. For a trip $-j$, the trip time (TT_j) is computed by substracting arrival and departure times of the trip. The total trip time (TTT) is then obtained by summing over all trips. To compute average travel speed (v), we calculate the total trip distance (TTD) and then divide it by total trip time.

$$TTT = \sum_j TT_j \tag{6}$$

$$TTD = \sum_j TD_j \tag{7}$$

$$v = \frac{TTD}{TTT} \tag{8}$$

where, TT_j and TD_j are trip time and trip length for a trip$-j$. Such a quantification makes every kilometer traveled by vehicles count and equitable.

Transport emission. The baseline transport emissions data were derived from the Climate Watch Database, which reported China's total transport-related emissions as 996.35 million tonnes (Mt) in 2021. Given that road transport contributes approximately 73.46% to global transport emissions, China's road transport emissions were estimated to be 731.89 Mt annually. To estimate city-specific emissions, we apportioned this total according to urban population size, assuming that per capita emissions within cities align with national averages.

To evaluate the impact of adaptive traffic signals on transport emissions, we utilized the Motor Vehicle Emission Simulator (MOVES), an on-road emission model developed by the United States Environmental Protection Agency. The MOVES model is recognized for its application in research and policy[46–48] due to its vehicle-specific power (VSP) framework, which enables detailed emission analyses. In this framework, VSP is calculated for each vehicle at a microscopic level based on instantaneous speed, acceleration, road grade, and road load coefficient. The VSP is computed using the following equation:

$$VSP = \frac{A \cdot v_t + B \cdot v_t^2 + C \cdot v_t^3 + m \cdot v_t \cdot a_t}{m} \tag{9}$$

where v_t is vehicle's speed at time t (m/s), a_t is vehicle's acceleration at time t (m/s^2), and m denotes vehicle mass (tonnes); A, B, C correspond to coefficients of rolling resistance, rotational resistance, and aerodynamic drag, respectively, and vary by vehicle type. For example, for a passenger 39 car, $A = 0.1565$, $B = 2.002 \times 10^{-3}$, $C = 4.926 \times 10^{-4}$, and $m = 1.479$ tonne. Although road grade is also a factor in VSP, default MOVES assume zero grade.

Using vehicle trajectory data from CBEngine, we classified vehicle driving modes based on instantaneous speed and calculated VSP values. Emissions for each vehicle were quantified according to emission rates specific to each driving mode (see Supplementary Table 1). The total emissions (TE) across all vehicle trips were calculated as follows:

$$E_{i,t} = f_i(VSP_t, v_t) \tag{10}$$

$$TE = \sum_i \sum_t E_{i,t} \tag{11}$$

where $E_{i,t}$ represents the emissions of vehicle i during time period t. Adaptive traffic signals influence vehicle driving modes by minimizing idle and braking times at intersections, which the MOVES model accounts for, enabling us to estimate CO_2 emission reductions attributable to these adaptive traffic signals.

Adaptive traffic signals can influence vehicle driving modes by reducing idle time and minimizing braking at intersections. By comparing emissions data under pretimed and adaptive traffic signal conditions, we assessed the reduction in CO_2 emissions attributable to adaptive traffic signals. Using MOVES, we estimated the average emissions change per vehicle due to adaptive signaling and scaled this difference to represent city-level changes, accounting for the city's population size. This approach allowed us to estimate the total CO_2 emissions reduction achievable through city-wide implementation of adaptive traffic signals. Fuel consumption was estimated on a gasoline basis. Using data from The Engineering ToolBox, which indicates that the combustion of 1 kg of gasoline produces approximately 3.3 kg of CO_2, we calculated reductions in gasoline consumption by multiplying CO_2 emission reductions by a factor of 0.30.

To estimate the number and proportion of petrol vehicles that need to be replaced by electric vehicles, we first use data from *Our World in Data*[49] to calculate the difference in CO_2 emissions per passenger-kilometer between petrol and electric vehicles. Next, we apply the annual average mileage of vehicles in China[50] to determine the annual emissions reduction when a vehicle is replaced by an electric engine. Finally, we divide the total emissions reduction achieved through adaptive signals by this annual per-vehicle reduction to estimate the number of vehicles that need to be replaced each year. The percentage is then calculated by comparing this figure with the total number of privately owned vehicles in China in 2022[51].

Implementation cost and societal benefit. The implementation cost was estimated based on 145 field projects recorded in the Intelligent Transportation Systems Joint Program. The implementation cost includes one-time installation cost and annual operation and maintenance cost. The per-intersection installation cost varies between US $22,000 and US$82,300, with an average of US$48,069. The annual operation and maintenance cost is estimated at US$1079 per intersection. The lifespan of the whole adaptive traffic signal control system was estimated to be about 71 months on average[36]. Therefore, the annual per-intersection implementation cost is US$9203, including both installation and operation costs.

The trip time benefit was estimated by investigating travelers' willingness to pay for travel time savings (i.e., value of trip time). The results from a stated preference (SP) survey conducted in the city of Nanjing in 2018 were adopted[38]. According to the survey, the mean estimated value of trip time is about US$4.84 (or 30.39 CNY) per hour. The trip time benefit was then estimated by multiplying the value of trip time with total trip time savings during both peak and off-peak hours.

We adopted the results from a study that used a global atmospheric model to simulate the benefits of global GHG reductions on air quality and human health[52]. According to the study, monetized emission benefit estimates are on average US$455 per ton of CO_2 for China. Gasoline was used as a surrogate for estimating benefits from fuel consumption savings. The price of gasoline was retrieved from Trading Economics, which is US$1,646 per ton (retrieved on 2022/01/20). The emission and fuel consumption benefits were estimated by multiplying the benefit per unit by total reductions of CO_2 emission and gasoline consumption, respectively.

Limitations

This study assesses the effectiveness of adaptive traffic signal control in mitigating urban congestion and transport emissions. The analysis is limited to China's top 100 congested cities (ranked by Gaode) and covers peak hours from 6:00 to 23:00 daily, excluding times and locations with minimal congestion. This restriction likely has minimal impact on our results, as adaptive and pre-timed signals perform similarly under uncongested conditions (e.g., speed changes under 1% in Urumqi and Yili). Trip time benefits were based on 2018 value of travel time saving data, and baseline BAU CO_2 emissions were calculated using 2021 data, assuming stable socioeconomic conditions. While this likely underestimates benefits due to China's economic growth and increased travel demand, the results still substantiate the cost-effectiveness of adaptive signals in congestion mitigation.

Increased speeds from adaptive signals may induce additional travel, as people opt to drive more or travel farther, potentially offsetting some congestion benefits. Our models do not fully capture induced demand due to data limitations, but adaptive signaling generally supports higher traffic volumes and smoother flows. Complementary measures, like improved public transit[53,54], could curb induced demand by encouraging shifts to sustainable modes. Further research is necessary to evaluate the impacts of adaptive signals on non-motorized users, such as pedestrians and cyclists.

Despite these limitations, this study provides insights into the role of adaptive traffic signals in reducing congestion and emissions, underscoring a cost-effective approach to urban carbon reduction. Our findings emphasize that adaptive signaling and other data-driven methods present viable pathways for cities seeking to address congestion while contributing to Sustainable Development Goals, particularly "sustainable cities and communities" and "climate action." This work also highlights the need for integrated data-sharing systems to support adaptive signaling and other data-driven urban governance initiatives, reinforcing the importance of cross-sectoral data integration in sustainable city management.

Data availability

Data supporting the findings of this study are available within the article (see "Data sources") and our Zenodo repository at https://doi.org/10.5281/zenodo.14591154.

Code availability

We use CBEngine for simulating city-scale traffic in this study. CBEngine is freely accessible by users and can be downloaded at https://github.com/CityBrainLab/CityBrainLab. The code for this research is available at https://github.com/RafaDD/BigData-TSC.

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Acknowledgements

This study was supported by Hangzhou Social Science Planning Project (Grant No. 24JD066, K.W.), the Key R&D Project of Science Technology Department of Zhejiang Province (Grant No. 2022C01121, K.W.), Zhejiang Provincial Department of Transport research projects (Grant No. ZJXL-JTT-202223, K.W.), the National Natural Science Foundation of China (42325707 and 42261144001, B.G.), the National Key Research and Development Project of China (2022YFE0138200, B.G.), and Frontiers Planet Prize Award: International Champion Prize funded by the Frontiers Research Foundation (B.G.).

Author contributions

B.G., K.W., F.J., and G.Z. designed the study. T.X., Y.S., and G.Z. processed the raw data and contributed to discussion of the paper. K.W., J.D., and J.L. conducted the research. K.W. wrote the first draft of the paper, B.G., Y.S., Y.Z, and T.X. revised the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-025-56701-4.

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Peer review information *Nature Communications* thanks the anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

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